

WASP-52b

R-0528

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_180703 · created 2026-07-21T05:43:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458302.8746 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.063
Transit depth (Rp/Rs)^2	3.32 +/- 2.3 [%]
Orbital Inclination (inc)	82.56 +/- 2.85
Airmass coefficient 1 (a1)	0.54 +/- 0.37
Airmass coefficient 2 (a2)	0.32 +/- 0.15
Scatter in the residuals of the lightcurve fit is	61.66 %
Best Comparison Star	#3 - [322, 135]
Optimal Aperture	2.76
Optimal Annulus	5.75
Transit Duration (day)	0.047 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.063 vs 0.1646 (deviation 0.26σ)
Duration fit vs published	0.047 d vs 0.0758 d (deviation 0.84σ)
Mid-time O–C	9.7 min
Depth SNR	2.7
Residual scatter	61.66 %

Light curve

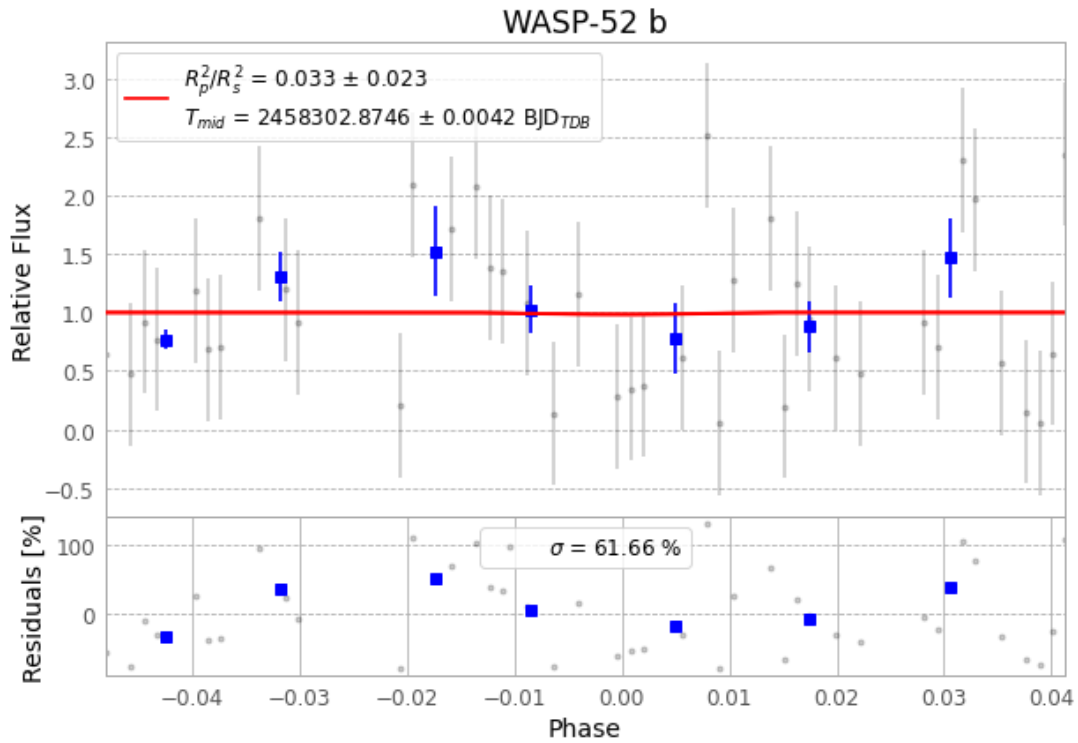


Figure 1 — FinalLightCurve_WASP-52 b_2018-07-02.png (SHA-256 5c0994f9ade9b4ee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 144], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 efaef790b6711c5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

77 FITS frames (51 MB), WASP-52180703065412.FITS → WASP-52180703104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180703.log (d51a3864dc54...), FinalLightCurve_WASP-52 b_2018-07-02.png (5c0994f9ade9...), FinalLightCurve_WASP-52 b_2018-07-02.csv (084c84f5dc73...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 05:43Z.